# Mid Formula

## (1) Floating Point

- Standard form / General / Convention 1 : $\pm (0.d_1 d_2 \ldots d_n)_\beta \times \beta^e$

  $d_1 = 1$ and $d_1 \neq 0$

- Normalized form : $\pm (0.1\, d_1 d_2 \ldots d_n)_\beta \times \beta^e$
- Denormalized form : $\pm (1.d_1 d_2 \ldots d_n)_\beta \times \beta^e$

## (2) Exponent biasing .

- Standard form : $\pm (0.d_1 d_2 \ldots d_{52})_\beta \times \beta^{e-1023}$
- Normalized form : $\pm (0.1 d_1 d_2 \ldots d_{51})_\beta \times \beta^{e-1023+1}$
- Denormalized form : $\pm (1.d_1 d_2 \ldots d_{52})_\beta \times \beta^{e-1023}$

## (3) Variations / কয়টা হতে পারে :

- Standard : $2^{m-1} \times \text{Sign} \times e$
- Normalized : $2^m \times \text{Sign} \times e$
- Denormalized : $2^m \times \text{Sign} \times e$

CamScanner

Lecture 2

## Rounding error:

Rounding Error $= |fl(x) - x|$

Scale invariant rounding error $= \dfrac{|fl(x) - x|}{|x|}$

$[\delta]$

Maximum scale invariant R.E / Machine epsilon/

Upper bound error / Unit Roundoff / Max Delta, $\delta_{max} = \dfrac{|fl(x) - x|}{|x|}$

↳ standard $= \frac{1}{2} \beta^{1-m}$

↳ Normalized $= \frac{1}{2} \beta^{-m}$

↳ Denormalized $= \frac{1}{2} \beta^{-m}$

## Taylor Series: * Derivation

$$f(x) = \underbrace{f(x_0)}_{\text{coeff}} + \underbrace{f'(x_0)}_{\text{coeff}}(x - x_0) + \underbrace{\frac{f''(x_0)}{2!}}_{\text{coeff}}(x - x_0)^2 + \underbrace{\frac{f'''(x_0)}{3!}}_{\text{coeff}}(x - x_0)^3$$

$$\text{Taylor error} = \frac{f^{n+1}(\varepsilon)}{(n+1)!}(x - x_0)^{n+1}$$

## Vandermond:

- $Va = f$
- $a = V^{-1} f$

↙ coeff ↘ Vandermond matrix

Problem:
* Expensive
* Not accurate

Solution: Use Lagrange

## Lagrange

$$P_n(x) = l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2) + \dots ,$$

$$l_k(x) = \frac{x - x_j}{x_k - x_j} \times \dots$$

Problem:
* Can not add new term

Solution: Use Newton D & D

## Newton's Divide and difference:

*Derivation

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots$$

↓ coeff

## Interpolation error / Cauchy Theorem / Truncation / Upper bound :

$$\left| \frac{f^{n+1}(\epsilon)\,(x-x_0)(x-x_1)\cdots(x-x_n)}{(n+1)!} \right|$$

$$= \underbrace{\left| \frac{f^{n+1}(\epsilon)}{(n+1)!} \right|}_{\epsilon \text{ part}} \times \underbrace{\left| (x-x_0)(x-x_1)\cdots(x-x_n) \right|}_{w(x) \text{ part}}$$

## Chebyshev Theorem :

$$A = \frac{b-a}{2} \quad [\text{Radius}]$$

$$\phi_j = \frac{(2j+1)\pi}{2(n+1)} \quad [\text{Angled point}]$$

$$x_j = A \cos \frac{(2j+1)\pi}{2(n+1)}$$

## Hermite Interpolation :

- $P_{2n+1} = h_0(x) f(x_0) + h_1(x) f(x_1) + h_2(x) f(x_2) + \cdots$
  $\hat{h}_0 f'(x_0) + \hat{h}_1 f'(x_1) + \hat{h}_2 f'(x_2) + \cdots$

- $h_k(x) = \{1 - 2(x-x_k) l_k'(x_k)\} l_k(x)^2$
- $\hat{h}_k = (x-x_k) l_k(x)^2$

## Differentiation:

- Forward = $\frac{f(x_0+h) - f(x_0)}{h} + \left| \frac{f''(\xi)(-h)}{2!} \right|$

  Forward ↑ Truncation error ↑

- Central = $\frac{f(x_0+h) - f(x_0-h)}{2h} + \left| \frac{f^3(\xi)\, h^2}{3!} \right|$

  central ↑ Truncation ↑

- Backward = $\frac{f(x_0) - f(x_0-h)}{h} + \left| \frac{f^2(\xi) h}{2!} \right|$

## Richardson: * Derivation

- $D_h' = \frac{4D_{h/2}^0 - D_h^0}{3}$

- $D_h^0 = \frac{f(x_0+h) - f(x_0-h)}{2h}$ [central]

- $D_{h/2}^0 = \frac{f(x_0+\frac{h}{2}) - f(x_0-\frac{h}{2})}{2\frac{h}{2}}$